Projectile Motion

Every final formula from Lectures 11, 12 and 13 on one sheet. Copy it into your notes and keep it in front of you while you solve.

THE MASTER METHOD

- Resolve into two axes
- Solve each as a 1D problem
- Join them through the shared t

Use this whenever a formula's assumption breaks. Every result on this sheet is just these three steps, worked out once.

SYMBOLS
u = launch speed · g = 10 m/s²

PARTS 1 AND 2
θ = launch angle, from the horizontal

PART 3 ONLY
θ = incline angle · α = launch angle, from the incline

Watch the angle names. In Parts 1 and 2, θ is measured from the **horizontal**. In Part 3 it is the **incline** angle, and α is the launch angle measured from the **incline surface**. Getting these two the wrong way round is the single most common mistake.

PART 1 GROUND TO GROUND

Launch and landing at the SAME height. Launch u at θ above the horizontal.

QUANTITY	FORMULA	NOTE
Components	$u_x = u\cos\theta \qquad u_y = u\sin\theta$	$a_x = 0,\ \ a_y = -g$
Time of flight	$T = \dfrac{2u\sin\theta}{g}$	from y = 0
Time to the top	$t = \dfrac{u\sin\theta}{g} = \dfrac{T}{2}$	up-time = down-time
Maximum height	$H = \dfrac{u^2\sin^2\theta}{2g}$	at the top $v_y = 0$
Range	$R = \dfrac{u^2\sin 2\theta}{g}$	$R = u_x \times T$
Maximum range	$R_{\max} = \dfrac{u^2}{g}$ at $\theta = 45^\circ$	$\sin 2\theta = 1$
Complementary angles	θ and $(90^\circ - \theta)$ give the same R	different H and T
Trajectory	$y = x\tan\theta - \dfrac{gx^2}{2u^2\cos^2\theta}$	a parabola
Velocity at time t	$v_x = u\cos\theta \qquad v_y = u\sin\theta - gt$	v_x never changes
Speed at the top	$v = u\cos\theta$	minimum speed, NOT zero
Useful relation	$R = 4H\cot\theta \ \Rightarrow\ \tan\theta = \dfrac{4H}{R}$	

PART 2A FROM A HEIGHT · LAUNCHED HORIZONTALLY

Launched from height h with $\theta = 0$. Part 1 formulas do NOT apply.

QUANTITY	FORMULA	NOTE
Components	$u_x = u \qquad u_y = 0$	free fall vertically
Time to land	$t = \sqrt{\dfrac{2h}{g}}$	independent of u
Range	$R = u\sqrt{\dfrac{2h}{g}}$	$R = u \times t$
Velocity on landing	$v_x = u \qquad v_y = gt = \sqrt{2gh}$	
Impact speed	$v = \sqrt{u^2 + 2gh}$	
Impact angle	$\tan\phi = \dfrac{\sqrt{2gh}}{u}$	below the horizontal

PART 2B FROM A HEIGHT · LAUNCHED AT AN ANGLE

Launched at θ above the horizontal from height h. Lands h BELOW the launch point.

QUANTITY	FORMULA	NOTE
Sign convention	$y_{\text{landing}} = -h$	UP = +, origin at the launch point
y-equation	$-h = u\sin\theta\, t - \frac{1}{2}gt^2$	a genuine quadratic
Time of flight	$t = \dfrac{u\sin\theta + \sqrt{u^2\sin^2\theta + 2gh}}{g}$	POSITIVE root only
Range	$R = (u\cos\theta)\, t$	
Height above launch	$H = \dfrac{u^2\sin^2\theta}{2g}$	above ground: h + H
Velocity on landing	$v_x = u\cos\theta \qquad v_y = u\sin\theta - gt$	v_x unchanged
Impact speed	$v = \sqrt{u^2 + 2gh}$	true for ANY θ
Impact angle	$\tan\phi = \dfrac{\lvert v_y\rvert}{v_x}$	below the horizontal

THE TEMPLATE FOR EVERY HEIGHT PROBLEM

1. **Sign convention** UP positive, origin at the launch point
2. **Landing value** y = −h, signed correctly
3. **Solve for t** from the y-equation
4. **Keep one root** the positive one only
5. **Get the range** $R = u_x \times t$

PART 3 ON AN INCLINED PLANE

Rotate the axes: x' ALONG the incline, y' PERPENDICULAR to it. Here θ is the INCLINE angle and α the launch angle from the incline surface.

QUANTITY	FORMULA	NOTE
Components	$u_{x'} = u\cos\alpha \qquad u_{y'} = u\sin\alpha$	α is from the incline
Acceleration $\perp$ to the slope	$a_{y'} = -g\cos\theta$	ALWAYS, both cases
Acceleration along the slope	$a_{x'} = \mp g\sin\theta$	– up the slope, + down
Time of flight	$T = \dfrac{2u\sin\alpha}{g\cos\theta}$	SAME for both cases
Max distance from the surface	$H_\perp = \dfrac{u^2\sin^2\alpha}{2g\cos\theta}$	
Range UP the incline	$R = (u\cos\alpha)T - \frac{1}{2}(g\sin\theta)T^2$	$= \dfrac{2u^2\sin\alpha\cos(\alpha+\theta)}{g\cos^2\theta}$
Range DOWN the incline	$R = (u\cos\alpha)T + \frac{1}{2}(g\sin\theta)T^2$	$= \dfrac{2u^2\sin\alpha\cos(\alpha-\theta)}{g\cos^2\theta}$
Velocity at time t	$v_{x'} = u\cos\alpha \mp (g\sin\theta)t \qquad v_{y'} = u\sin\alpha - (g\cos\theta)t$	– up, + down
Best angle UP the incline	$\alpha_{\text{opt}} = \dfrac{90^\circ - \theta}{2} \qquad R_{\max} = \dfrac{u^2}{g(1+\sin\theta)}$	
Best angle DOWN the incline	$\alpha_{\text{opt}} = \dfrac{90^\circ + \theta}{2} \qquad R_{\max} = \dfrac{u^2}{g(1-\sin\theta)}$	
Check	$\theta = 0 \;\Rightarrow\; \alpha_{\text{opt}} = 45^\circ,\; R_{\max} = \dfrac{u^2}{g}$	Part 1 is the special case

THE THREE CASES AT A GLANCE

	GROUND TO GROUND	FROM A HEIGHT	ON AN INCLINE
Landing condition	$y = 0$	$y = -h$	$y' = 0$
Find t by	factorising	the quadratic formula	$y' = 0$
Time of flight	$\frac{2u\sin\theta}{g}$	$\frac{u\sin\theta + \sqrt{u^2\sin^2\theta + 2gh}}{g}$	$\frac{2u\sin\alpha}{g\cos\theta}$
Range	u_xT	u_xt	$u\cos\alpha\, T \mp \frac{1}{2}g\sin\theta\, T^2$
Best angle	45°	–	$\frac{90^\circ \mp \theta}{2}$